Fish-pond Change Detection Based on Short Term Time Series of RADARSAT Images and Object-oriented Method

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***Abstract*— As an important type of wetlands, fish-pond provides a number of important ecosystem services. It was easily affected by human activities such as the change of cultivation structure. Tracking the fast change of fish-pond can obtain the information that the environment is impacted by human activities to what extent. In this study, we proposed a change detection procedure in order to delineate fast change fish-pond using high resolution, short term time series of SAR imagery. The procedure includes (1) SAR images preprocessing, (2) object extraction, (3) multi-temporal change characteristics analysis of fish-pond, (4) change objects classification, and (5) a final change results analysis. In this study, object-oriented method was adopted to extract and analyze the texture characteristics of fish-pond in the high spatial resolution of SAR data. Objects were generated with suitable scale parameters by means of multi-temporal segmentation approach according to the image features (including backscatter, difference and ratio of three temporal RADARSAT images, shape characters), and the changed objects were extracted through analyzing the shape index and contextual information generated by Definiens software. Six change types of fish-pond were achieved. The results indicate that high resolution SAR data can provide better information for fish-pond change detection in the study area, and the procedure proposed in this study can efficiently obtain the fast change objects with high accuracy between three short term time series SAR images, which are useful for the further classification.**

Keywords- SAR; short term time series, change detection; fish-pond; wetlands

I. INTRODUCTION

Change detection can identify and quantify temporal differences between the states of an object or phenomenon [1-2]. It is an important methodology to detect the changes that occur on the Earth surface with multi-temporal remote sensing images. This depends on the fact that the knowledge of the dynamics of either natural resources or man-made structures is a valuable source of information for the human being [3]. A variety of efficient digital change detection techniques has been developed in the past two or three decades, which can be divided into about four main types, that is, algebra approach (differencing, rationing, and regression) [4-8], change vector analysis approach [9-11], transforming (including PCA, multivariate alteration detection, and chi-square transformation) [12-15], post-classification comparison approach (including supervised and unsupervised classification methods) [16-17].

Fish-pond, as one type of wetlands, can provide a number of important ecosystem services, such as flood control and water storage, recharge of groundwater, and adjusting local climate et al [18-19]. However, fish-ponds are usually developed into other farm lands by local farmers for their valuable economic value. During the change process, the ecosystem value of fish-ponds is generally overlooked. Tracking fish-pond change using short term time series remote sensing images can obtain the information that the environment is impacted by human activities to what extent.

High spectral, high spatial resolution optical images can afford detailed land use and land cover information for change detection classification and analysis. Optical images have been demonstrated as the most important data sources in many change detection analyses. However, they still have posed problems to the application field for the imaging conditions influence, especially the atmospheric and weather condition [20]. It is difficult to acquire the high-quality optical images on the date of every satellite passing, in south china. Unlike optical sensors, synthetic aperture radar (SAR) can penetrate through the cloud, they have the ability to collect high quality images under the complex weather conditions, which can provide efficient data source for fast change detection analysis.

Conventional methodologies mainly centralize in seeking change information based on difference between the pixel classifications. This is time consuming and easily results in misjudgment. While object-oriented approach can well integrates the spectral feature with the texture, shape and contextual information for classification. The objects consisting of many pixels are firstly achieved by means of image segmentation based on spectral feature and shape characters [21]. The change objects can then be efficiently identified through the classification contrast. Compared with other land cover types, fish-ponds have strong texture and shape characteristics; they generally show as inerratic objects in remote sensing images, which is very suitable for object-oriented classification. The change information of fish-ponds can be extracted according to shape information, and neighborhood relationships analysis.

In this study, we proposed a change detection procedure in order to delineate fast change fish-pond using high resolution, short term time series of SAR imagery. The procedure includes (1) SAR image preprocessing, (2) object extraction, (2) multi-temporal change characteristics analysis of fish-pond, (4) change objects classification, and (5) a final change results analysis. Object-oriented method was adopted to extract and analyze the texture characteristics of fish-pond in multi-temporal SAR images. Objects were generated with suitable scale parameters by means of multi-temporal segmentation approach according to the image features (including backscatter, difference and ratio of images), and the changed objects were extracted through analyzing the shape and contextual information generated by Definiens software. The procedure proposed in this study was further validated with three short term RADARSAT images. Six types of change objects with high accuracy were obtained in the sample area.

II. METHODOLOGY

Object-based procedure was used for detecting and classifying fish-pond changes with short term time series of SAR imagery. The procedure mainly consists of five steps: first step concerns SAR imager preprocessing (including geometric correction, co-registration, and noise reduction). Second step relates to object extraction using multi-temporal segmentation approach. Third step deals with multi-temporal change characteristics analysis of fish-pond. Fourth step describes the classification of change objects. Then, the last step aims at final analysis of change results. The flowchart of the procedure is shown as Figure 1. The steps above are explained in details as follows.

A. SAR images preprocessing

Generally, the SAR data is saved as SLC format (single look complex). To provide the image features for object segmentation, The SAR data should be preprocessed and transformed through the following steps. Firstly, the SLC images should be converted to the magnitude images by radiometric calibration. Radiometric calibration of the SAR data was carried out using the Radar Analysis Package of the PCI software. Calculating the backscattering value of each pixel involves three procedures: (a) converting the pixel DN values into the slant range backscattering coefficient (Beta Nought) using the scaling and gain coefficients in the raw data, (b) calculating the incidence angle for each pixel of the images in the range direction, (c) incorporating the incidence angle to extract the backscatter coefficient (Sigma Nought) from the Beta Nought values, and (4) converting the 16-bit data to unsigned 8-bit images.

Secondly, multi-temporal magnitude images should be registered with the same spatial reference. Image registration was then implemented using the AutoSync component of the ERDAS software. This component provides the Automatic Point Measurement (AMP) tool for finding the control points, which are used to register these SAR images. A second order polynomial model was applied by transforming these SAR images. The average RMS error was less than 1 pixel for the transformation.

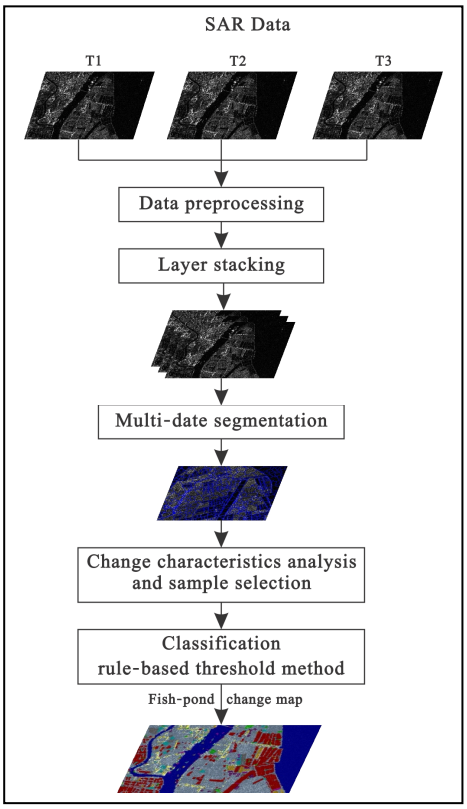

Figure 1. Flowchart of the change detection procedure proposed in this study

As is well known, SAR images are often affected by noise called speckle. Therefore, speckle reduction techniques were employed to minimize this noise effect before carrying out image segmentation. The gamma map filter (GMF) was used to remove high-frequency noise (speckle) while preserving high-frequency features (edges) for SAR images. Through shape adaptive windowing, GMF detectors allow the use of large window sizes for better speckle reduction while preserving spatial resolution and structural features. Following speckle reduction, these SAR data were resampled to 6 * 6m.

B. Object extraction (multi-date segmentation)

Image segmentation is firstly carried out to extract objects and provide information for classification. Meaningful image objects based on image spectral and textural characteristics were extracted in this process [22]. The object-oriented analysis in this study has been carried out with a bottom-up region merging technique in Definiens 7.0 software. Where, adjacent image pixels are merged into one image object, and the image objects are connected together through evaluating their heterogeneity.

The three important parameters are considered by the segmentation algorithm embedded in Definiens: scale, color (spectral information), and shape. Color and shape parameters can be weighted from 0 to 1; and shape is also divided into two setting, smoothness and compactness, which can be complementarily weighted from 0 to 1 [25]. Scale is the key parameter used to control the average image objects size [26]. The process stops when the smallest object growth exceeds the threshold defined by the scale parameter [23-24]. Once the

appropriate scale parameter is identified, the boundary and shape of the image objects can be refined with the color and shape parameters. Most published works have found that more meaningful objects are extracted with a higher weight for the color criterion [27-28].

The multi-date segmentation techniques are adopted in this study [15], image objects are extract from a stacking of multi-spectral images. A 3-dimension file including three temporal SAR images was input to the segmentation procedure. Objects boundaries were delimitated according to spectral properties, spatial context characteristics, and the temporal behavior of the pixels.

C. Change Characteristics analysis

Each object extracted from the temporal segmentation has two optional land cover types, that is, fish-pond or non-fish-pond. As for three temporal SAR images, there will have eight status for every object ($2^3 = 8$), which is listed in Table I (1 is fish-pond and 2 is non-fish-pond or other land covers).

TABLE I. EIGHT TYPE OF CHANGE STATUS OF FISH-POND IN THIS STUDY

T1	T2	T3	Code	Change/Un-change
1	1	1	111	Un-change
1	1	2	112	Change
1	2	1	121	Change
1	2	2	122	Change
2	1	1	211	Change
2	1	2	212	Change
2	2	1	221	Change
2	2	2	222	Un-change

There are two un-changed types and six changed types of fish-pond in three temporal images. If the time interval between the three images is short, the change types can demonstrate the fast change characteristics of fish-pond. In this study, the changed types were further concerned with classification and change detection technique.

D. Classification of change objects

The rule-based approach is used to classify and indentify the changed objects in this study. The thematic information or land covers types were obtained from classification rules. The following features determined the expression of the rules, such as backscatter of three temporal SAR images, shape features (density, length/width, shape index, and roundness).These can be calculated in Definiens.

As is well known, image difference between two temporal images is the most popular features used for change detection analysis. Although there are many other complex methods suitable for change detection, image difference approach was still employed due to the complexity of change types in this study. Rather than concerning whether the objects or pixels between two temporal images changed or not, the status of fish-pond would be identified into different types in this study. Using the features of difference between the images can reduce the computation complexity and improve the classification accuracy.

A set of classification rules will be used to identify the six types of change status. Classification rules are normally expressed as the “IF condition, THEN action” statements. The condition portion is usually set as a threshold criterion for one or more classification features, and the action portion is usually a decision process that objects meeting the condition portion will be assigned into a single change status. The threshold criterions for classification rules were automatically generated in Definiens 7.0 according to samples or training data.

E. Change analysis

After completing the classification procedure, a series of statistical analysis with the change objects were carry out in this study. The number and the total area of change objects for each change status were calculated, and the final change maps will be provided after accuracy assessment.

III. CASE STUDY

A. SAR data

In this study, the three temporal Radarsat-1 C band HH polarization (horizontal-transmit and horizontal-receive) images were used to character fish-pond change detection in the study area. SAR data were acquired on January 27, 2006, February 20, 2006, and March 16, 2006. The interval of two temporal is less than one month. Figure shows the SAR data (acquired on January 27, 2006) for this study. The raw SAR images had a resolution of 4.6 * 5.1 m and a swath width of 50 km on the ground. The incidence angle was about 45。

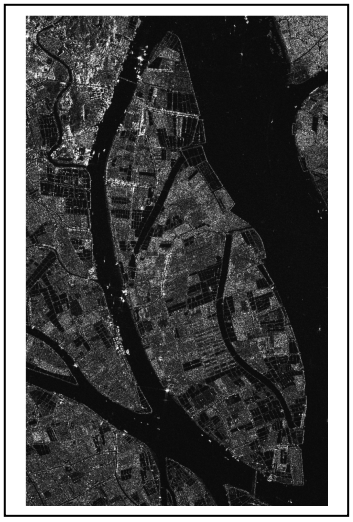

Figure 2. SAR image used for this study

B. Software

Definiens, (formally known as eCognitaion), is developed from the Munich Definiens company, which was founded by the Nobel laureate Gerd Binning [29]. Definiens is the first and the most popular commercial software for object-oriented analysis with many segmentation algorithms.

C. Classification

To obtain the small homogeneous objects for best delineate the change types of fish-pond, the multi-date segmentation procedure was performed with the optimal parameters (Scale = 10, color = 0.7, shape = 0.3, smoothness = 0.5, and compactness = 0.5) according to a series of experiments. A total of 15129 objects were produced.

About 200 sample objects were selected as training data according to field investigation and visual selecting on other high resolution optical images acquired in the same time period. Based on these training data, the backscatter characteristics of some land cover types were analyzed. The backscatter of river, fish-pond, other land, and residential land are scattered within the range of 1-2, 2-5, 10-20, and 25-255, respectively.

The difference images were calculated between time 1 and time 2, time 2 and time 3, which was represented as $D_{t1\text{-}t2}$ and $D_{t2\text{-}t3}$ in this study. Training objects have such features such as backscatter of three temporal, a series of shape index, and difference information.

All training objects were input into Definiens for generating the thresholds and selection classification features with nearest neighbor approach. We developed a set of classification rules in the process tree window in Definiens. All the 11 types of land cover, including six change status and other five land covers, were classified, and each land cover type can be defined as a few classification rules.

The ships in the river were also classified and defined as some changed objects surrounding by river objects. This will be further processed by means of noise filter approach. Figure 3 shows the final classification map.

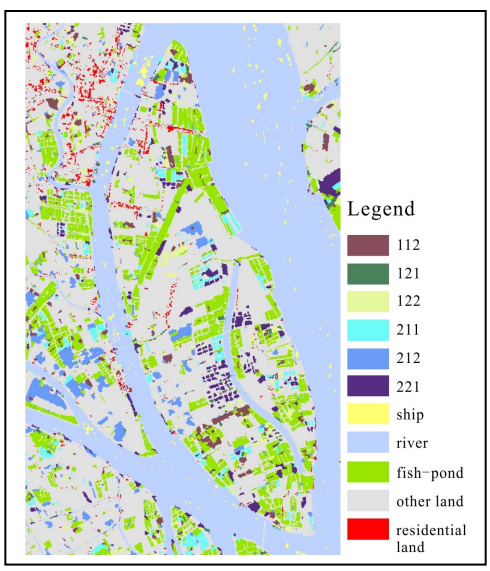

Figure 3. The fast change maps of fish-pond in three temporal

D. Accuracy assessment

Other 196 sample objects were selected as test dataset for accuracy assessment procedure. Accuracy assessment was also performed in Definiens 7.0. These sample objects were created to a TTA Mask image used for generating error matrix and kappa statistics. Table II shows the error matrix, overall accuracy, and kappa statistics for this classification.

TABLE II. ERROR MATRIX OF FISH-POND CHANGE DETECTION

types	112	121	122	211	212	221	LA	RE	RI	FP	Sum
112	18	0	1	0	1	0	0	0	0	0	20
121	0	17	1	0	0	0	0	0	0	1	19
122	0	1	13	1	1	1	0	0	0	0	17
211	0	0	0	16	0	0	2	0	0	0	18
212	0	0	0	0	17	0	0	0	0	0	17
221	0	1	0	1	0	19	0	0	0	0	21
LA	0	0	1	1	1	0	14	4	0	0	21
RE	0	0	0	0	0	0	0	18	0	0	18
RI	0	0	0	0	0	0	0	0	12	0	12
FP	2	1	4	1	0	0	0	0	6	19	33
Sum	20	20	20	20	20	20	16	22	18	20	19
Total Accuracy: 83.2%; Kappa: 0.81											

LA= other lands; RE= residential land; RI = river; FP =fish-pond.

The overall accuracy shows as83.1% and kappa is 0.81. It is demonstrated that the procedure is efficient for the fast change detection of fish-pond.

E. Results and discussion

Also the statistics analysis was executed based on the classification map. The analysis results were shown in table III. The largest area of the changed status is 212, which represents that these objects in the three temporal experienced change process of un-fish-pond, fish-pond, and un-fish-pond, respectively.

TABLE III. CHANGE ANALYSIS OF FISH-POND IN THREE TEMPORAL

Change status	Number of objects	Area / hm^2	Portion of change to total land area
112	281	4.68	2.39%
121	121	1.41	0.72%
122	403	5.14	2.63%
211	345	5.69	2.90%
212	386	8.18	4.17%
221	387	6.39	3.26%

Total area of all the changed fish-pond is 31.49 hm^2, which is the 16.07% to total area of fish-pond. It is indicated that freshwater feeding is one of the main industry.

IV. CONCLUSION

In this study, a procedure based on object-oriented method was introduced to detect the fast changes of fish-pond with short temporal SAR data. The following conclusions can be drawn.

(1) It is confirmed that High resolution Radarsat SAR data are suitable for identifying the change of fish-pond. The backscatter and texture characteristics shown between the images with short time interval can provide efficient information for fast change detection.

(2) This object-based method is easily repeatable and can be directly performed in the Definiens software environment. With this method, Change information is classified and indentified from the homogeneous image objects *in this study*.

The change detection results indicated that object-based method is very available for this classification.

(3) The procedure proposed here proved to be very efficient to identify fish-pond fast changes. A detection accuracy higher than 83% and an overall kappa higher than 0.80 were achieved using a SAR multi-date data covering no more than three months.

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